

Aim: Create a database having three tables to store the details of students of Computer Department in your college. Personal_Information (Roll_No, Name, DOB, Address, Marks (rounded to a whole number) in percentage at 10 +2, phone_number) Paper_Details (Paper code, Name of the Paper) Student's Academic and Attendance details (College roll number, Paper code, Attendance, Marks in home examination).

```
-- Create Database
CREATE DATABASE CollegeDB;
USE CollegeDB;
```

1) Identify primary and foreign keys.

Ans. Primary Key: Roll_No, Paper_Code, (College_Roll_No, Paper_Code).

Foreign Keys: College_Roll_No, Paper_Code.

2) Create the tables and insert at least 5 records in each table.

Input:

```
-- 1. Personal_Information Table
CREATE TABLE Personal_Information (
    Roll_No INT PRIMARY KEY,
    Name VARCHAR(100),
    DOB DATE,
    Address VARCHAR(200),
    Marks_12th INT CHECK (Marks_12th BETWEEN 0 AND 100), -- rounded %
    Phone_Number VARCHAR(15)
);
-- Insert 5 records in Personal_Information
INSERT INTO Personal_Information VALUES
(101, 'Amit Sharma', '2002-05-12', 'Delhi', 85, '9876543210'),
(102, 'Riya Gupta', '2001-08-22', 'Mumbai', 90, '9876501234'),
(103, 'Karan Mehta', '2002-11-05', 'Delhi', 78, '8765432109'),
(104, 'Pooja Singh', '2001-03-15', 'Kolkata', 82, '7654321098'),
(105, 'Arjun Verma', '2002-01-25', 'Delhi', 88, '6543210987');
SELECT * FROM Personal_Information;
```

Output:

Roll_No	Name	DOB	Address	Marks_12th	Phone_Number
101	Amit Sharma	2002-05-12	Delhi	85	9876543210
102	Riya Gupta	2001-08-22	Mumbai	90	9876501234
103	Karan Mehta	2002-11-05	Delhi	78	8765432109
104	Pooja Singh	2001-03-15	Kolkata	82	7654321098
105	Arjun Verma	2002-01-25	Delhi	88	6543210987
NULL	NULL	NULL	NULL	NULL	NULL

Input:

```
-- 2. Paper_Details Table
CREATE TABLE Paper_Details (
    Paper_Code INT PRIMARY KEY,
    Paper_Name VARCHAR(100)
);
-- Insert 5 records in Paper_Details
INSERT INTO Paper_Details VALUES
(1, 'Data Structures'),
(2, 'Database Management Systems'),
(3, 'Operating Systems'),
(4, 'Computer Networks'),
(5, 'Artificial Intelligence');
SELECT * FROM Paper_Details;
```

Output:

Paper_Code	Paper_Name
1	Data Structures
2	Database Management Systems
3	Operating Systems
4	Computer Networks
5	Artificial Intelligence
NULL	NULL

Input:

```
-- 3. Academic_Attendance Table
CREATE TABLE Academic_Attendance (
    College_Roll_No INT,
    Paper_Code INT,
    Attendance DECIMAL(5,2), -- % attendance
    Marks_Home_Exam INT CHECK (Marks_Home_Exam BETWEEN 0 AND 100),
    PRIMARY KEY (College_Roll_No, Paper_Code),
    FOREIGN KEY (College_Roll_No) REFERENCES Personal_Information(Roll_No),
    FOREIGN KEY (Paper_Code) REFERENCES Paper_Details(Paper_Code)
);
-- Insert 5 records in Academic_Attendance
INSERT INTO Academic_Attendance VALUES
(101, 1, 80.5, 65),
(101, 2, 78.0, 70),
(102, 1, 60.0, 55),
(102, 2, 82.0, 65),
(103, 1, 90.0, 72),
(103, 2, 76.0, 40),
(104, 2, 88.0, 85),
(104, 3, 70.0, 60),
(105, 1, 92.0, 75),
(105, 2, 80.0, 68);
SELECT * FROM Academic_Attendance;
```

Output:

College_Roll_No	Paper_Code	Attendance	Marks_Home_Exam
102	2	82.00	65
103	1	90.00	72
103	2	76.00	40
104	2	88.00	85
104	3	70.00	60
105	1	92.00	75
105	2	80.00	68
NULL	NULL	NULL	NULL

- 3) Design a query that will return the records (from the second table) along with the name of student from the first table, related to students who have more than 75% attendance and more than 60% marks in paper 2.

Input:

```
SELECT pi.Name, pd.Paper_Name, aa.Attendance, aa.Marks_Home_Exam
FROM Personal_Information pi
JOIN Academic_Attendance aa ON pi.Roll_No = aa.College_Roll_No
JOIN Paper_Details pd ON aa.Paper_Code = pd.Paper_Code
WHERE aa.Paper_Code = 2
      AND aa.Attendance > 75
      AND aa.Marks_Home_Exam > 60;
```

Output:

Name	Paper_Name	Attendance	Marks_Home_Exam
Amit Sharma	Database Management Systems	78.00	70
Riya Gupta	Database Management Systems	82.00	65
Pooja Singh	Database Management Systems	88.00	85
Arjun Verma	Database Management Systems	80.00	68

- 4) List all students who live in “Delhi” and have marks greater than 60 in paper 1.

Input:

```
SELECT pi.Name, pi.Address, aa.Marks_Home_Exam
FROM Personal_Information pi
JOIN Academic_Attendance aa ON pi.Roll_No = aa.College_Roll_No
WHERE pi.Address = 'Delhi'
      AND aa.Paper_Code = 1
      AND aa.Marks_Home_Exam > 60;
```

Output:

Name	Address	Marks_Home_Exam
Amit Sharma	Delhi	65
Karan Mehta	Delhi	72
Arjun Verma	Delhi	75

- 5) Find the total attendance and total marks obtained by each student.

Input:

```
SELECT pi.Name,  
       SUM(aa.Attendance) AS Total_Attendance,  
       SUM(aa.Marks_Home_Exam) AS Total_Marks  
FROM Personal_Information pi  
JOIN Academic_Attendance aa ON pi.Roll_No = aa.College_Roll_No  
GROUP BY pi.Name;
```

Output:

Name	Total_Attendance	Total_Marks
Amit Sharma	158.50	135
Riya Gupta	142.00	120
Karan Mehta	166.00	112
Pooja Singh	158.00	145
Arjun Verma	172.00	143

- 6) List the name of student who has got the highest marks in paper 2.

Input:

```
SELECT pi.Name, aa.Marks_Home_Exam  
FROM Personal_Information pi  
JOIN Academic_Attendance aa ON pi.Roll_No = aa.College_Roll_No  
WHERE aa.Paper_Code = 2  
ORDER BY aa.Marks_Home_Exam DESC  
LIMIT 1;
```

Output:

Name	Marks_Home_Exam
Pooja Singh	85